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CLDV POE Reflective Report

CLDV POE

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1 EVENTEASE CLOUD APPLICATION

EventEase is a cloud-based event management application, created using ASP.NET Core MVC and Microsoft Azure services. The purpose of this application is to allow users to manage venues, events and bookings through a consolidated and centralized system. In the process of development, the application had been migrated from locally hosted resources (Azurite, SQL server) to Azure cloud services. Providing both practical experience with cloud deployments and cloud resource management.

2 APPLICATION FEATURES

The Application Includes a wide array of management and validation features that were designed to support event administration for the company's employees. Of which include:

Feature	Description
Venue Management	Create, view , edit and delete venues
Event Management	Create, view , edit and delete events
Booking Management	Create, view , edit and delete bookings
Image Uploads	Upload and store venue and event images
Search Functions	Search bookings using different types of search parameters
Filtering	Filter venues and events using specific parameters (event type, venue availability)
Validation	Prevent invalid actions and duplicate bookings

3 MANAGEMENT SYSTEMS

3.1 Venue Management

The venue page allows booking specialists to maintain different venue records, whereby each venue contains information such as venue name, location capacity, availability and an image of the venue itself.

3.2 Event Management

The event page allows users to create and manage upcoming events. Each event contains an event name, description, image, event type, and an image. The event types are stored in a lookup table to make sure that event listing remains consistent throughout the registration process.

3.3 Booking Management

The booking page acts as a junction between events and venues, Users can assign events to venues while also registering specialist information and booking dates,

4 SEARCH AND FILTERING

The following search and filtering features were implemented:

Functionality	Feature
Booking Search	Search By: Booking ID, Event Name, Booking Date
Event Filtering	Filter By: Event Type, Event Date range
Venue Filtering	Check: Available option

5 IMAGE MANAGEMENT

Images are uploaded, through the applications' interface and is then stored in Azure blob storage as compared to its previous container azurite(which stores images locally). Instead of storing images files directly into the database only the image file name is stored in the Azure SQL Database. This helped in reducing database sizes and improves scalability.

6 VALIDATION AND ERROR HANDLING

These were the validation mechanisms that were implemented to improve the applications security.

Mechanism	Description
Duplicate Booking	The system checks whether a venue already has an event scheduled during the selected period. If an overlap in event time exists, the booking system rejects the entry, and an error message is displayed.
Deletion Protection	The system works to prevent deletion of events and venues when associated bookings exist. It helps to prevent instances of orphaned records.
File Validation	Only image formats are accepted (.jpg, .jpeg, .png) are processed.
Venue Availability	Venue marked "Unavailable" cannot be selected for a new booking. This helps to make sure users can't schedule events that cannot be accommodated for.

7 AZURE SERVICES USED

Azure Service	Intended Purpose
Resource Group	Used to organize all the project resources
Azure SQL Database	Stores the applications' relative data
Azure Storage Account	Provides cloud storage
Blob Container	Acts as storage for uploaded images
Azure App Service	Used to host web applications

8 AZURE RESOURCES

8.1 Azure SQL Database

Azure SQL Database was used to host the application data that includes events, venues bookings, and event types. Using Azure SQL help in allowing for real-time updates to be reflected in the application's codebase without the need to re-instantiate a database connection.

8.2 Azure Blob Storage

Azure blob storage was used to aid in storing the uploaded images. This separates the file storage from the database storage, minimizing the amount of database resources used.

8.3 Azure App Service

Azure App service was used to host the EventEase application, making the web app accessible through a web browser that would not need local execution.

9 THE MIGRATION EXPERIENCE

The most important parts of this development journey were the migration from local resources to Azure cloud resources and services. Whereby, initially, the application was developed with the use of SQL Server Management Studio (localhost) to act as a local SQL database for the application data, and Azurite to act as the application's local blob storage emulator. These would help with app testing and development without the need for cloud resources.

9.1 Migration to Azure SQL Database

The application had been migrated from local DB to Azure SQL Database; the following changes had been made:

Configuration Changes

Changes Made	How they were implemented
Updated connection strings	Connection strings were switched out from that of the local DB with the Azure SQL DB
Firewall Rules	Azure SQL Firewall rules were configured to make use of all of the Azure resources and allow device connection to the website.
Database Authentication settings	Settings were configured to accommodate connections between the website and the Azure SQL database.
App Settings	Application settings were re-configured to accommodate cloud connections.

Challenges Experienced

- **Firewall Configuration issue** – During testing the application could not be executed due to the Azure cloud service blocking the application from running online. It wasn't until the firewall rules were configured to accommodate the test devices ip address before the application would run smoothly again.
- **Connection String Errors** – During the initial setup of the newly switched out connection strings, an error occurred where the application still ran off of the

local DB connection strings, a re-configured appsettings.json file with the appropriate connection strings helped mitigate this.

9.2 Migration From Azurite to Blob Storage

The image storage functionality was migrated from Azurite to Azure Blob Storage.

To make this transition happened the following changes were implemented:

Changes	Implementation
Storage Account	Storage account had been created.
Blob Container	Blob container had been created.
Connection Strings	App settings were updated to accommodate the blob containers connection strings.
Upload verification	Verification of image upload functionalities were enabled.

This transition helped in making sure the application could access cloud resources more efficiently, improve its overall reliability as an application that works over the internet instead of locally and provides greater scalability based on the instances of users and interactions that occur when using the application.

10 THE TECHNOLOGIES USED

Technology	Intended Purpose
C#, SQL	Backend development
ASP.NET Core MVC	Web Application Framework
Entity Framework Core (EF Core)	Integration with the database
SQL Server	Used to set up relational database
Azure SQL Database	Enable cloud database hosting
Azure Blob Storage	Image storage container
Bootstrap Framework	Styling the user interface
HTML, CSS	Setting up page structure and user interface design.

The development of the EventEase web application provided both practical experience in cloud deployment and web app development. The migration from local resources to Azure illustrated how development and production environments differ while also emphasizing the overall benefits that come with cloud hosted infrastructures that make web app deployments easier, though services that implement validation, storage solutions, and cloud services, all of which helped in the development of this functional event management system.